SANKALP TIWARI

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WORK EXPERIENCE

Software Engineer, Intercontinental Exchange — Atlanta, GA, USA **February 2026 – Present**

- Architected an **AI-enabled engineering platform** integrating LLM-driven generation, enterprise data retrieval, and deterministic validation, adopted by **100+ engineers** and reducing workflow authoring effort by up to **90%**.
- Built **LLM-assisted workflow generation pipelines** in Python with structured outputs and automated validation and repair loops, reducing manual authoring and review effort by approximately **70%** across multiple clearing applications.
- Engineered **high-throughput financial data-processing workflows** using streaming and batched processing for large reference-data payloads, reducing peak memory utilization by **65%** and eliminating legacy processing bottlenecks.
- Developed reusable **backend and data-integration services** across **multiple clearing applications** supporting positions, risk, pricing, settlement, and reference-data workflows.

Software Engineer 2, Goldman Sachs — New York, NY, USA **October 2024 – December 2025**

- Enhanced ingestion reliability by building **Spring Boot microservices** on internal Kubernetes clusters to process **8.5M+ financial events per day** from 6+ upstream systems via Kafka, persisting normalized data in MongoDB.
- Increased accuracy of intraday multi-region payment calculations by creating **event-processing services** in Spring Boot that validated and computed **70,000+ position events per day** with automated rule checks.
- Enhanced scalability and **reduced downstream lag by nearly 40%** by onboarding 20+ settlement and reporting services to internal Kafka topics handling **5,000-10,000 messages per minute**.
- Built a **real-time audit logging system** in Spring Boot and MongoDB to centralize high-volume processing events and accelerate production incident investigation.
- Cut reconciliation time by **50%** by building an **LLM-powered reconciliation system** grounded in internal documentation and business rules, automatically processing **270K+ daily announcements** and generating claims calculations and reporting.
- **Reduced exception volume by 90%** and improved **SLA compliance by 40%** by designing a new exception workflow in Spring Boot that automated triage of 3,000+ monthly exceptions.
- Improved **batch-processing throughput by 35%** by optimizing 20+ Autosys jobs, reducing average runtime from **18 mins to 11 mins**.
- Automated **CI/CD pipelines** with Jenkins and GitLab for Kubernetes-based microservices, improving deployment reliability and release velocity.

Software Developer & Compliance Support, SJSU Research Foundation — San Jose, CA, USA **May 2023 – September 2024**

- Improved system performance by deploying a **React-based Faculty Info System** on Ubuntu VMware for **500+ faculty members**.
- Built a React/Node.js form workflow integrated with university portals through REST APIs, improving compliance operations.
- Centralized application data in SQL and migrated legacy records to GCP, consolidating infrastructure and improving retrieval performance.

Software Engineer, Stetig Consulting — Pune, Maharashtra, India **June 2020 – June 2022**

- Reduced case-resolution time by **84%** with an **NLP-based chatbot microservice** integrated with backend APIs.
- **Increased client conversion by 23%** by building a **Spring Boot microservice** that generated personalized PDFs using Apache PDFBox.
- Developed real-time **Kafka pipelines** and modernized legacy SOAP services to REST, improving partner-system reliability.
- Built scalable REST APIs using **API Gateway, AWS Auto Scaling, and load balancers** for high-traffic applications.

SKILLS

Languages: Java, Python, Go, TypeScript/JavaScript, SQL, C++, Shell
Applied AI: LLM Applications, Agentic Workflows, RAG, Structured Outputs, LLM Evaluation, Machine Learning
Frameworks & Web: Spring Boot, FastAPI, Flask, Node.js, Express.js, React, Next.js, Redux, HTML, CSS, REST APIs, GraphQL
Distributed Systems & Data: Kafka, Kubernetes, Docker, Redis, MongoDB, PostgreSQL, MySQL, Sybase, Stored Procedures
Cloud & DevOps: AWS, GCP, Jenkins, GitLab CI/CD, SonarQube, Autosys, Git

PROJECTS

Moodify | GitHub — React, Node.js, Flask, AWS, Spotify API, TMDB

- Built a mood-based music and movie recommendation app using ML (Random Forest) and cosine similarity for content filtering.

EDUCATION

Master of Science in Software Engineering, San Jose State University, *San Jose, CA, USA* **Aug 2022 – May 2024**
Bachelor of Engineering in Information Technology, RBU (formerly RCOEM), *Nagpur, MH, India* **Aug 2016 – May 2020**